Dynamic Pricing Optimizer

A Machine Learning Project Report

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1. Introduction

Dynamic pricing is a widely used pricing strategy in e-commerce where product prices are adjusted according to market conditions such as demand, competitor pricing, seasonality, and promotions. Unlike fixed pricing, dynamic pricing helps businesses respond to changing conditions and improve revenue generation.

This project, titled **Dynamic Pricing Optimizer**, is an AI-powered pricing analytics system that predicts demand-related behaviour and estimates the optimal selling price for a product. The system uses machine learning models and a Streamlit-based dashboard to provide real-time pricing insights through an interactive web interface.

The main goal of this project is to simulate a real-world e-commerce pricing environment and identify the price point that maximizes revenue while considering business constraints such as product cost and promotional activity.

2. Problem Statement

In e-commerce, selecting the right product price is a difficult decision. If the price is set too high, demand may drop significantly. If the price is too low, the company may lose profit even if sales volume increases. Therefore, the challenge is to find a price that balances demand and profitability.

The problem addressed in this project is:

1. Predict product demand or units sold based on market and business factors.
2. Evaluate multiple regression models and select the best performing one.
3. Use the selected model to estimate revenue for different price points.
4. Determine the optimal price that maximizes revenue.

This project demonstrates how machine learning can support pricing decisions in online retail systems.

3. Dataset Description

The dataset used in this project contains structured e-commerce pricing data. Each row represents a pricing scenario with market and business features. The main columns are:

- **price**: current selling price of the product
- **competitor_price**: competitor's product price
- **cost_price**: cost of the product
- **promo_flag**: whether a promotion is active
- **day_of_week**: day number in the week
- **is_weekend**: weekend indicator
- **month**: month of the year
- **units_sold**: number of units sold, used as a demand-related target

The dataset is suitable for regression-based modeling because the output values are continuous. The features capture both market competition and temporal effects, which are important in dynamic pricing.

4. Methodology

The project follows a supervised machine learning workflow:

4.1 Data Preprocessing

The data is first cleaned and prepared for training. Missing values, if any, are handled appropriately. Categorical values such as promotion status and weekend indicator are converted into numerical form so that machine learning models can process them.

4.2 Feature Selection

The following variables are used as input features:

- competitor_price
- cost_price
- promo_flag
- day_of_week
- is_weekend
- month
- units_sold

The target variable is the **price**.

4.3 Model Training

Instead of relying on a single algorithm, multiple regression models are trained and compared:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor

These models are evaluated using regression metrics on the test set. The final model is selected based on the best performance, especially the lowest error. The code follows a model-comparison approach rather than using only one model.

4.4 Model Selection

After training all candidate models, the system compares their test performance and chooses the best one. This makes the project stronger because the final prediction model is chosen empirically instead of manually.

5. Price Optimization Logic

After the best model is selected, the application evaluates different price points and computes the expected revenue for each one.

The revenue formula used is:

Revenue = Price × Units Sold

The system estimates units sold under different pricing scenarios and finds the price that gives the maximum revenue. This value is displayed as the **optimized price**.

This approach makes the project more useful than a simple prediction model because it directly supports business decision-making.

6. Web Application

A web interface is built using **Streamlit** to make the system interactive and easy to use. The dashboard allows users to enter:

- Current Price
- Competitor Price
- Cost Price
- Promotional Status
- Day of Week
- Weekend Indicator
- Month

After receiving the inputs, the application:

1. Loads the trained model
2. Predicts expected units sold
3. Calculates expected revenue
4. Identifies the optimal price
5. Plots a **Revenue vs. Price** curve

The dashboard provides a practical way to test multiple pricing scenarios in real time.

7. Results and Discussion

The system successfully demonstrates how machine learning can be used for pricing optimization in e-commerce. By comparing several regression models and selecting the best performing one, the project improves prediction reliability.

The Revenue vs. Price curve shows an important real-world behavior: revenue increases with price up to a certain point and then starts decreasing as demand falls. The optimal price lies near the peak of this curve.

This confirms the usefulness of dynamic pricing and price optimization in business settings.

8. Conclusion

This project presents a machine learning-based dynamic pricing system that helps determine the optimal product price for revenue maximization. The system uses structured e-commerce data, compares multiple regression models, and deploys the best model in a Streamlit dashboard.

The project demonstrates the complete pipeline of data preparation, model comparison, optimization, and visualization. It can be further improved by adding profit maximization, demand forecasting, and live competitor price scraping.

9. Future Scope

Possible future improvements include:

- Profit-based optimization instead of only revenue maximization
- More advanced time-series demand forecasting
- Live competitor price extraction using web scraping
- Automated retraining with new sales data
- Cloud deployment for production use